

NETWORK LAYER (OSI LAYER 3)

The network layer provides services that allow end devices to exchange data across networks. Its main protocols are IPv4 and IPv6.

OTHER NETWORK LAYER PROTOCOLS

- Routing Protocols – OSPF
- Messaging Protocols – ICMP

THE OSI MODEL



NETWORK LAYER PROTOCOLS

- Internet Protocol version 4 (IPv4)
- Internet Protocol version 6 (IPv6)



NETWORK LAYER – FOUR BASIC OPERATIONS

- 1. Addressing End Devices**
 - Every end device must have a unique IP address for identification on the network.
- 2. Encapsulation**
 - The network layer adds an IP header to the transport layer data (PDU) to form an IP packet.
 - Includes source and destination IP addresses.
- 3. Routing**
 - Directs packets to a destination on another network.
 - Routers choose the best path to reach the destination.
 - A packet may pass through many routers.
 - Each router a packet crosses is called a hop.
- 4. De-encapsulation**
 - At the destination, the IP header is checked.
 - If the destination IP matches, the header is removed and the data (Layer 4 PDU) is passed to the appropriate transport layer service.

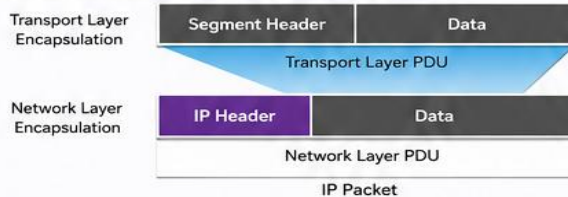


Unlike the transport layer (Layer 4), which manages data between processes on each host, the network layer (IPv4 & IPv6) defines packet structure and processing to deliver data from one host to another. It carries packets for multiple types of communications between multiple hosts.

8.1.2 IP ENCAPSULATION

IP adds an IP header to the transport layer PDU to deliver the packet to the destination host.

Result: Network layer PDU becomes an IP packet.



The IP header is examined by routers (Layer 3 devices) as the packet travels across the network. The data (transport layer PDU) remains unchanged during this process.

8.1.3 CHARACTERISTICS OF IP

- 1. Connectionless**
No connection is established before sending data packets.
- 2. Best Effort**
Packet delivery is not guaranteed. Packets may be lost, duplicated, delayed or arrive out of order.
- 3. Media Independent**
Operation does not depend on the medium (copper, fiber-optic, wireless, etc.).

Note NAT (Network Address Translation) is discussed in later modules.

8.1.4 – 8.1.6 CONNECTIONLESS, BEST EFFORT, MEDIA INDEPENDENT

8.1.4 CONNECTIONLESS

No dedicated end-to-end connection is created before data is sent.

Analogy (Mail)



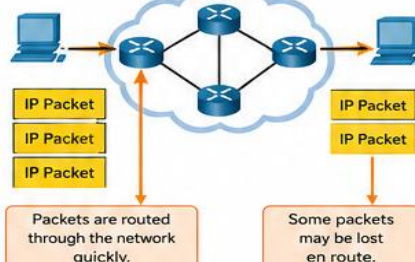
Network Example



IP requires no initial exchange of control information to establish an end-to-end connection.

8.1.5 BEST EFFORT

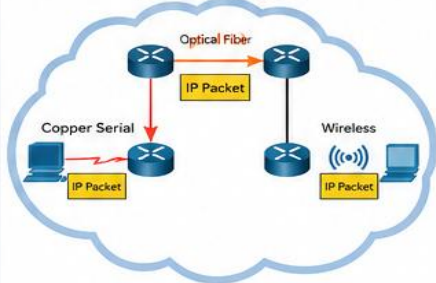
IP does not guarantee that all sent packets will be received.



Unreliable or "best-effort" delivery – packets may be lost, duplicated, delayed, or arrive out of order.

8.1.6 MEDIA INDEPENDENT

IP operates independently of the media that carries the data.



IP packets can travel over different media (copper cable, fiber, serial link, wireless).

KEY TAKEAWAYS



Network layer = OSI Layer 3.



IPv4 and IPv6 are the principal protocols.



Provides addressing, encapsulation, routing and de-encapsulation.



IP is connectionless, best effort and media independent.



Routers forward packets based on IP addresses to reach the destination.

IPv4 PACKET HEADER

8.2.1 WHAT IS IPv4 PACKET HEADER?



IPv4 is one of the primary network layer communication protocols.



The IPv4 packet header ensures that a packet is delivered to its next stop on the way to its final destination device.



An IPv4 packet header contains fields with important information about the packet. These fields hold binary values that are examined by the Layer 3 (Network Layer) process.

8.2.2 IPv4 PACKET HEADER FIELDS



The binary values of each field define how the IP packet is handled.

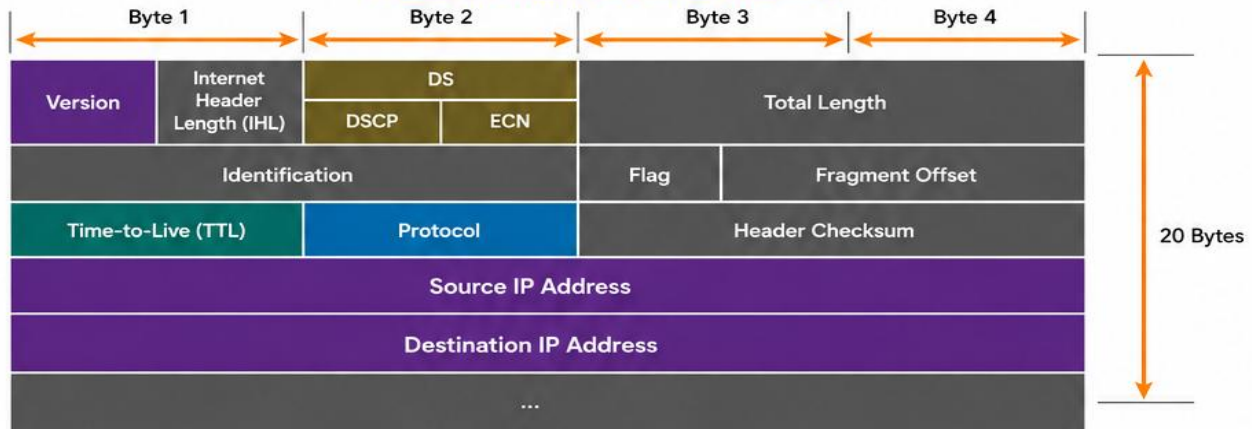


Protocol header diagrams, read left to right and top to bottom, help us understand the roles of each field.



The diagram below shows the fields in an IPv4 packet header.

FIELDS IN THE IPv4 PACKET HEADER



SIGNIFICANT FIELDS EXPLAINED

Version (4 bits)	Indicates the IP version. For IPv4, the value is 4.
Internet Header Length (IHL) (4 bits)	Specifies the length of the IP header in 32-bit words (4 bytes). Minimum value is 5 (20 bytes with no options).
DS (8 bits)	Type of Service (ToS). Contains two sub-fields: DSCP (6 bits): Used for Differentiated Services (prioritization of packets). ECN (2 bits): Used for Explicit Congestion Notification (ECN).
Total Length (16 bits)	Total length of the IPv4 packet (header + data) in bytes. Maximum value is 65,535 bytes.
Identification (16 bits)	Identifies fragments of the same original packet. Used during fragmentation and reassembly.
Flags (3 bits)	Used for fragmentation control. Helps indicate whether more fragments follow.
Fragment Offset (13 bits)	Shows the position of this fragment in the original data. Helps in reassembling the packet at the destination.
Time-to-Live (TTL) (8 bits)	Limits how long the packet can live in the network. Decremented by each router. When it reaches 0, the packet is discarded.
Protocol (8 bits)	Identifies the next upper layer protocol carried in the data portion. Examples: ICMP (1), TCP (6), UDP (17).
Header Checksum (16 bits)	Used by the receiver to verify if the header is correct. If error is detected, the packet is discarded.
Source IP Address (32 bits)	IP address of the sender (32-bit IPv4 address).
Destination IP Address (32 bits)	IP address of the intended receiver (32-bit IPv4 address).
Options and Padding (variable)	Optional fields for special purposes. Rarely used and beyond the scope of this module.

KEY POINTS



The most commonly referenced fields are the **Source** and **Destination IP Addresses**.



These fields show where the packet is coming from and where it is going.



Internet Header Length, Total Length, and Header Checksum help routers process the packet correctly.



Other fields (Identification, Flags, Fragment Offset) are used for fragmentation and reassembly when packets are split.



Options and Padding fields are rarely used and are beyond the scope of this module.



Note: During transmission from source to destination, most fields remain the same. Routers mainly update the Time-to-Live (TTL) and Header Checksum.

QUICK TAKEAWAY



IPv4 header = 20 bytes (minimum). Carries control information for delivery.



Contains 13 main fields that help in addressing, routing, error checking, and fragmentation.



Routers read header fields to forward the packet to the next best path.



Ensures packets reach their next hop safely until they reach the final destination.

IPv6 PACKET – LIMITATIONS OF IPv4 & OVERVIEW

1. LIMITATIONS OF IPv4

Even though IPv4 is still in use, it has some major issues which led to the development of IPv6.



1. IPv4 Address Depletion

- Limited number of unique public addresses.
- Only about 4 billion addresses.
- Growing number of devices needs more addresses.



2. Lack of End-to-End Connectivity

- NAT is used so multiple devices can share one public IPv4 address.
- Real public address of internal hosts is hidden.
- Can be problematic for technologies that require direct end-to-end connectivity.



3. Increased Network Complexity

- NAT was only a transition mechanism.
- NAT adds extra complexity, latency and makes troubleshooting difficult.

2. IPv6 OVERVIEW

IPv6 was developed by the IETF in the 1990s to replace IPv4.

Key Improvements of IPv6

- ✓ Increased address space (128-bit addressing)
- ✓ Simpler header (fewer fields)
- ✓ Eliminates the need for NAT
- ✓ Better suited for future network demands



IPv4 Address Space

- 32-bit addresses
- $\approx 4,294,967,296$ addresses (≈ 4 billion)

VS

IPv6 Address Space

- 128-bit addresses
- 340,282,366,920,938,463,463,374,607,431,768,211,456 (≈ 340 undecillion)

This is roughly equivalent to every grain of sand on Earth!



IPv4 AND IPv6 ADDRESS SPACE COMPARISON

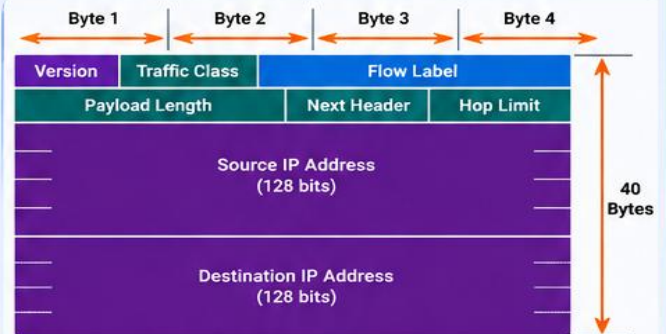
Number Name	Scientific Notation	Number of Zeros
1 Thousand	10^3	1,000
1 Million	10^6	1,000,000
1 Billion	10^9	1,000,000,000
1 Trillion	10^{12}	1,000,000,000,000
1 Quadrillion	10^{15}	1,000,000,000,000,000
1 Quintillion	10^{18}	1,000,000,000,000,000,000
1 Sextillion	10^{21}	1,000,000,000,000,000,000,000
1 Septillion	10^{24}	1,000,000,000,000,000,000,000,000
1 Octillion	10^{27}	1,000,000,000,000,000,000,000,000,000
1 Nonillion	10^{30}	1,000,000,000,000,000,000,000,000,000,000
1 Decillion	10^{33}	1,000,000,000,000,000,000,000,000,000,000,000
1 Undecillion	10^{36}	1,000,000,000,000,000,000,000,000,000,000,000,000

Legend

= 4 billion IPv4 addresses

= 340 undecillion IPv6 addresses

FIELDS IN THE IPv6 PACKET HEADER



HEADER FIELDS EXPLAINED

Version (4 bits)	Value = 6, identifies this as an IPv6 packet.
Traffic Class (8 bits)	• Similar to IPv4 DS field (Quality of Service).
Flow Label (20 bits)	• All packets with the same label receive same handling by routers.
Payload Length (16 bits)	• Length of the payload (data only). Does NOT include the 40-byte header.
Next Header (8 bits)	• Replaces IPv4 Protocol field. Identifies the next upper-layer protocol.
Hop Limit (8 bits)	• Similar to TTL. Decrement by 1 at each router. Packet discarded when it reaches 0.
Source IPv6 Address (128 bits)	IPv6 address of the sender.
Destination IPv6 Address (128 bits)	IPv6 address of the receiver.



Extension Headers (EH)

- Optional headers in IPv6 packet.
- Placed between IPv6 header and payload.
- Used for fragmentation, security, mobility and more.

3. IPv4 vs IPv6 – KEY DIFFERENCES

	IPv4	IPv6
Address Size	✓ 32-bit (≈ 4 billion)	✓ 128-bit (≈ 340 undecillion)
Header Size	✓ Min 20 bytes	✓ Fixed 40 bytes
Header Fields	✓ Many fields	✓ Simpler (fewer fields)
NAT	✓ Required (due to address shortage)	✓ Not required
End-to-End Connectivity	✓ Limited (due to NAT)	✓ Direct connectivity enabled
Fragmentation	✓ Routers may fragment	✓ Routers do NOT fragment
Security / Mobility	✓ Limited	✓ Better support (using extension headers)

QUICK TAKEAWAYS



IPv6 solves key limitations of IPv4.



Larger address space → No NAT needed.



Simpler header → Faster and easier processing.



Provides true end-to-end connectivity.



Better for current and future network demands!

HOST ROUTING & ROUTING TABLES (IPv4)

8.4.1 HOST FORWARDING DECISION

Packets are always created at the source host.
The source host must deliver the packet to the destination host.

Host end devices create a routing table to help the network layer decide where to send packets.

TYPES OF DESTINATION HOSTS



Itself (Loopback)

Send to 127.0.0.1 (IPv4) or ::1 (IPv6) called the loopback interface.



Local Host

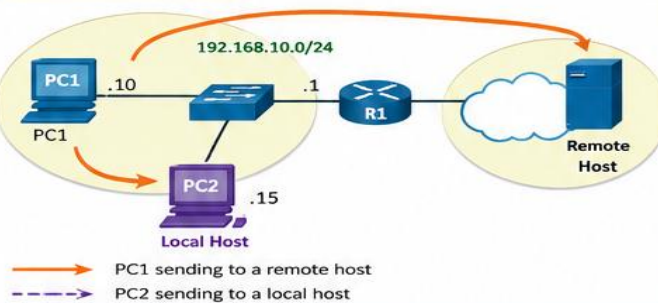
Destination is on the same local network as the sender. Source and destination share the same network address.



Remote Host

Destination is on a different network. Source and destination do not share the same network address.

EXAMPLE TOPOLOGY



8.4.2 DEFAULT GATEWAY

The default gateway is the router interface used by hosts to reach networks outside their local network.

A default gateway is used when:

- The destination IP is on a remote network.
- There is no specific route in the routing table.
- All traffic that is not local is sent to the default gateway.



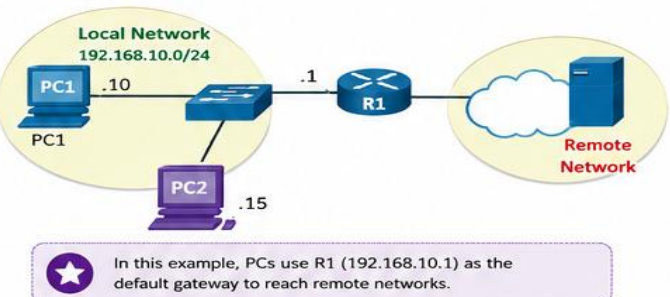
It acts as the exit path from your network to other networks (Internet, other offices, etc.).

8.4.3 HOST ROUTES TO THE DEFAULT GATEWAY

A host routing table includes a default gateway.

In IPv4, the next hop is the IPv4 address of the default gateway (either learned via DHCP or configured manually).

Example Topology



8.4.4 HOST ROUTING TABLES

On Windows, the commands **route print** or **netstat -r** display the host routing table.

- Both commands show the same output.
- netstat -r** output may look complex at first, but it is simple once understood.

Sample Topology



netstat -r (output omitted)

```
C:\Users\PC1> netstat -r
IPv4 Route Table
=====
Active Routes:
Network Destination  Netmask          Gateway          Interface        Metric
0.0.0.0              0.0.0.0          192.168.10.1     192.168.10.10    25
127.0.0.0            255.0.0.0        On-link          127.0.0.1        306
127.0.0.1            255.255.255.255  On-link          127.0.0.1        306
127.255.255.255      255.255.255.255  On-link          127.0.0.1        306
192.168.10.0         255.255.255.0    On-link          192.168.10.10    281
192.168.10.10        255.255.255.255  On-link          192.168.10.10    281
192.168.10.255       255.255.255.255  On-link          192.168.10.10    281
224.0.0.0            240.0.0.0        On-link          192.168.10.10    281
224.0.0.0            240.0.0.0        On-link          192.168.10.10    281
255.255.255.255      255.255.255.255  On-link          127.0.0.1        306
255.255.255.255      255.255.255.255  On-link          192.168.10.10    281
(output omitted)
```

Note: The output only displays the IPv4 route table.

The **netstat -r** or **route print** output has three important sections:



Interface List

- Shows the MAC address and interface number of each network-capable interface (Ethernet, Wi-Fi, Bluetooth, etc.).



IPv4 Route Table

- Lists all known IPv4 routes including direct connections, local network routes and the local default route.



IPv6 Route Table

- Lists all known IPv6 routes including direct connections, local network routes and the local default route.

COMMON netstat -r FIELDS (IPv4)

Field	Meaning	Example / Notes	Field	Meaning	Example / Notes
Network Destination	The destination network or host.	0.0.0.0 = default route 127.0.0.0 = loopback	Metric	Cost to reach the destination. Lower value = preferred path.	25, 281, 306
Netmask	Subnet mask for the destination.	255.255.255.0 = /24	On-link	Destination is directly connected to this interface.	No gateway needed.
Gateway	Next hop router.	192.168.10.1 (default gateway)	0.0.0.0/0	Default route (send all unknown traffic to this gateway).	Used for remote networks
Interface	IP address of the interface used to reach destination.	192.168.10.10 (local IP)			

KEY TAKEAWAYS



Hosts use routing tables to decide where to send packets.



Default gateway sends traffic to remote networks.



netstat -r or **route print** shows the host routing table.



Routing tables contain direct routes, local routes, and default routes.



Routers identify the best path and forward packets to the next hop.

INTRODUCTION TO ROUTING (PART 1)

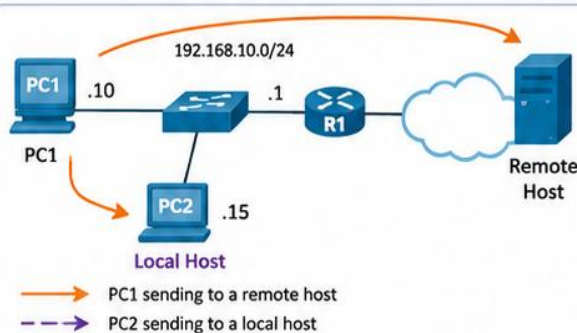
8.5.1 ROUTER PACKET FORWARDING DECISION

- Routers (intermediary devices) connect different networks.
- When a router receives a packet, it checks its routing table to decide where to send the packet next.
- These next hops/interfaces are called route entries.



What happens when a packet arrives at a router?

- 1 Router examines the destination IP address.**
 - Looks in the routing table for the best (longest matching) route entry.
- 2 Finds the next hop or exit interface.**
 - This tells the router where and how to forward the packet.
- 3 Forwards the packet.**
 - Sends the packet out of the selected interface toward the next hop or destination network.



R1 ROUTING TABLE (EXAMPLE)

Route (Network)	Next Hop or Exit Interface
192.168.10.0/24	G0/0/0
209.165.200.224/30	G0/0/1
10.1.1.0/24	via R2
Default Route 0.0.0.0/0	via R2



- The router uses the destination IP and routing table to find the next hop or exit interface.
- The next hop is the IP of the next router or the exit interface that leads toward the destination network.

8.5.2 IP ROUTER ROUTING TABLE

A router's routing table contains all possible network destinations and three types of route entries.



1. Directly-connected Networks

- Active router interfaces.
- Added when an interface with an IP address is up.
- Example in R1 table: 192.168.10.0/24



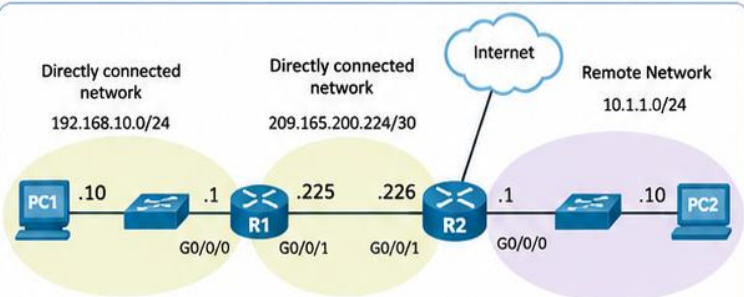
2. Remote Networks

- Networks not directly connected.
- Learned dynamically via routing protocols or configured by administrator.
- Example in R1 table: 10.1.1.0/24



3. Default Route

- Gateway of last resort.
- Used when no better (longer) match exists.
- Example in R1 table: 0.0.0.0/0



- R1 has two directly connected networks:
 - 192.168.10.0/24
 - 209.165.200.224/30
- R1 also has remote network (10.1.1.0/24) and the Internet.

Routers learn remote routes in two ways:



- **Manually** – Static routes are entered by administrator.



- **Dynamically** – Routes are learned automatically using a dynamic routing protocol.

SUMMARY: ROUTING TABLE ENTRIES

Type of Route	Description	Example in R1 Table
 Directly-connected Networks	Routes to networks the router is directly connected to via its interfaces.	192.168.10.0/24
 Remote Networks	Routes to networks not directly connected; learned dynamically or configured manually.	10.1.1.0/24
 Default Route (Gateway of last resort)	Used when no more specific route is found.	0.0.0.0/0



KEY
TAKEAWAY



Router receives
packet.



Checks destination IP
in routing table.



Finds best match
(longest match).

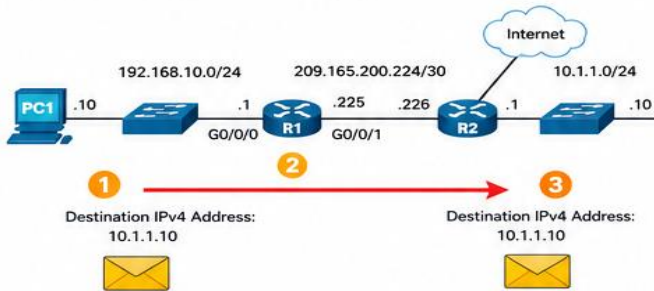


Forwards packet via
next hop/exit interface.

INTRODUCTION TO ROUTING (PART 2)

8.5.3 STATIC ROUTING

- Static routes are manually configured by the network administrator.
- Route entries include the remote network address and the IP address of the next hop router.

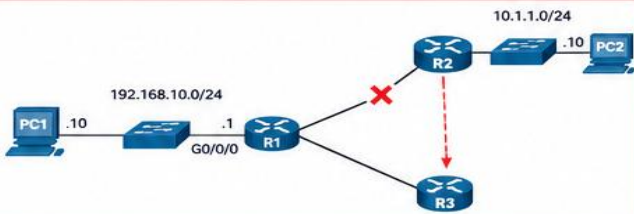


Static route example on R1:

```
R1(config)# ip route 10.1.1.0 255.255.255.0 209.165.200.226
```

Remote network address IP address of next hop router

If the topology changes, static routes do not update automatically. An admin must add new routes or remove/modify old ones. Example: If R1 loses the link to R2, a new static route via another router is required to reach 10.1.1.0/24.



R1 and R3 have higher priority routes, so routes are not automatically changed if the path to R2 goes down.

KEY FEATURES OF STATIC ROUTING

- Manually configured by administrator.
- Requires reconfiguration when topology changes.
- Best for small networks with few or no redundant links.

8.5.5 ROUTER PACKET FORWARDING PROCESS (RECAP)



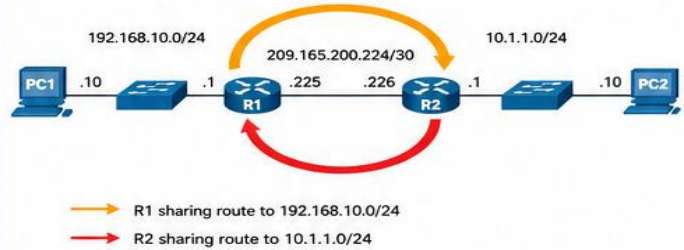
ROUTE ENTRY = NETWORK + NEXT HOP/INTERFACE

If multiple entries match, the longest (most specific) match is used.
If no match is found, the default route is used (if present).

The default gateway is the router interface used by hosts to reach networks outside their local network.

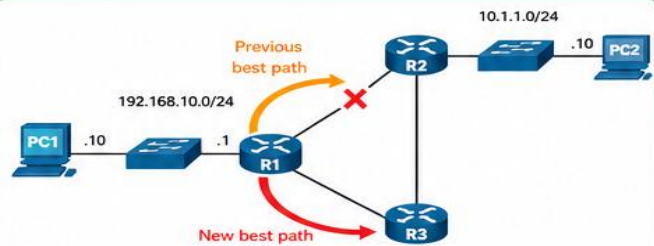
8.5.4 DYNAMIC ROUTING

- Dynamic routing protocols allow routers to automatically learn routes to remote networks, including a default route.
- Routers exchange route information with each other.
- Common protocols: OSPF, EIGRP.



Dynamic routing helps routers:

- Discover remote networks
- Maintain up-to-date routing information
- Choose the best path automatically
- Adapt to changes in the network



Routes are automatically updated when there is a change. Network continues to operate with minimal disruption.

Note: Dynamic routing is commonly used with extension headers and in large, changing networks.

8.5.6 INTRODUCTION TO AN IPv4 ROUTING TABLE

Example shows R2 connected to the Internet. R1 and R2 use OSPF to learn directly connected networks. R1 also has a default static route to reach other networks.

Sample: R1 Routing Table (show ip route)

```
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS level-1-2
       * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route
       m - application route
       + - replication route, % - next hop override, p - overrides from PFR

Gateway of last resort is 209.165.200.226 to network 0.0.0.0
S* 0.0.0.0/0 [1/0] via 209.165.200.226, GigabitEthernet0/0/1
C 10.1.1.0/24 is directly connected, GigabitEthernet0/0/1
O 192.168.10.0/24 [110/2] via 209.165.200.226, 00:00:10,
  GigabitEthernet0/0/1
O 209.165.200.224/30 is directly connected, GigabitEthernet0/0/1
C 209.165.200.226/32 is directly connected, GigabitEthernet0/0/1
```

Common Route Codes

Code	Meaning	Code	Meaning
L	Directly connected local interface IP address	IA	OSPF inter area route
C	Directly connected network	*	Candidate default route
S	Static route (manually configured)	U	Per-user static route
O	OSPF route	+	Replication route
D	EIGRP route	%	Next hop override